Fake News Detection
with Character-level CNN

Deniz Altıntaş [1]

1 FMV Işık University, Institute of Graduate Studies, Information Technologies, İstanbul, Türkiye; 24ITEC5008@gmail.com

Abstract

Fake news poses a serious challenge in online media by rapidly spreading misinformation and influencing public opinion. Traditional word-level text classification models may fail to capture subtle stylistic patterns present in deceptive content. This study investigates whether a Character-level Convolutional Neural Network (CharCNN) can improve fake news detection by learning directly from raw text.

Three models are evaluated: a TF-IDF-based Logistic Regression baseline, a hyperparameter-optimized variant, and a Character-level CNN. Experiments are conducted on the ISOT and WELFake datasets to assess both performance and cross-dataset generalization. The results show that while hyperparameter optimization improves baseline performance, the Character-level CNN achieves the highest accuracy and macro-F1 score. Error analysis further reveals that character-level modeling reduces misclassification in stylistically ambiguous and politically framed articles. These findings highlight the effectiveness of character-level approaches for robust fake news detection.

Keywords: Fake news detection, character-level CNN, text classification

1. Introduction

Can a Character-level Convolutional Neural Network (CharCNN) effectively detect fake news by learning from raw text? To address this question, this study compares a traditional TF-IDF-based baseline, an optimized variant, and a character-level CNN architecture.

Fake news has become a major challenge in online media influencing public opinion and spreading misinformation rapidly. This study aims to explore whether a Character-level CNN can capture these subtle linguistic signals to improve the accuracy and generalization of fake news detection models.

2. Related Work

Zhang et al. (2015) introduced the Character-level Convolutional Neural Network (CharCNN) a model that learns directly from raw character sequences without relying on word embeddings. This approach demonstrated strong performance in text classification by capturing subword and stylistic patterns.

Wang et al. (2020) proposed an N-gram-based CNN model for short text classification, improving local feature extraction through attention mechanisms, while Jang et al. (2020) developed a hybrid CNN–BiLSTM–Attention model that enhanced accuracy by integrating convolutional and sequential learning.

Liu et al. (2021) combined topic modeling (LDA) with transformer-based embeddings (ALBERT) and CNN layers for multi-label classification, and Chen et al. (2022) designed a BERT–CNN architecture for long-text news classification. Zhou (2022) proposed a TF-IDF + CNN-LSTM hybrid that optimized feature selection and reduced training time.

Despite these advancements few studies have applied character-level CNNs to fake news detection representing a significant gap that this research aims to address.

3. Methodology

3.1 Dataset

This project uses the ISOT Fake News Dataset created by the Information Security and Object Technology (ISOT) Lab at the University of Victoria [7]. The dataset contains approximately 36,000 news articles, evenly distributed between Fake and Real labels. Each entry includes the title and text of news articles collected from verified sources and unreliable websites.

The dataset is balanced, open-access and well-documented. For testing and validation, the WELFake Dataset from Kaggle will be used. It includes around 72,000 English news articles, preprocessed and labeled for fake news classification tasks [8].

Using two datasets allows for cross-dataset evaluation, testing whether the Character-level CNN can generalize across different data sources.

Other benchmark datasets, such as FakeNewsNet, will also be reviewed for future extensions or cross-comparison. [9]

Table 1. Dataset distribution

Dataset	Rows	Fake (%)	Real (%)
ISOT	39,103	45.8%	54.2%
WELFake	24,586	55%	45%
Total	63,689	49.5%	50.5%

3.2 Dataset Preprocessing

The study utilizes two publicly available fake news datasets, ISOT and WELFake. Both datasets contain a balanced distribution of Fake (0) and Real (1) labels, avoiding issues related to class imbalance. Before modeling, all text samples were cleaned through standard preprocessing steps, including lowercasing, removal of HTML artifacts, and normalization of punctuation and whitespace.

Prior to text cleaning, basic data quality checks were conducted for each dataset, including label distribution analysis, missing value inspection, duplicate text detection, and descriptive statistics of text length in terms of both characters and words. Records with missing combined text fields were removed, and duplicate news articles were eliminated to prevent information leakage and redundancy during training.

Text preprocessing was then applied using a two-stage cleaning pipeline. First, URLs, HTML tags, and formatting artifacts were removed, followed by whitespace normalization. After cleaning, additional auxiliary features such as word count, character count, and uppercase character ratio were extracted to support exploratory analysis and slice-based evaluation in later stages.

Although preprocessing steps such as null removal and duplicate filtering were applied, no change in dataset size was observed. This is because both datasets were already well-curated, containing no missing values and no duplicate text entries. Therefore, preprocessing mainly focused on text normalization and feature extraction rather than data reduction.

Table 2. Extracted auxiliary features after preprocessing

Feature	Description
clean_text	URLs and HTML removed, normalized text
word_count	Number of words after preprocessing
char_count	Number of characters after preprocessing
upper_ratio	Ratio of uppercase letters

3.3 Exploratory Data Analysis (EDA)

The dataset is nearly balanced between Fake and Real news, ensuring that model performance is not biased toward a dominant class.

Exploratory analysis revealed distinct structural and stylistic differences between the two classes. Real news articles are notably longer, averaging around 780 words, while fake news averages approximately 410 words.

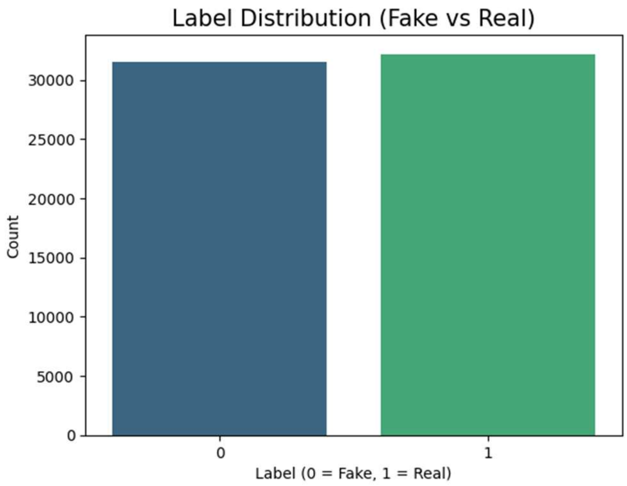

Figure 1. Label Distribution

Sentiment analysis further showed that fake news demonstrates more extreme polarity -either strongly positive or strongly negative- whereas real news tends to stay neutral. Frequency-based analyses (unigrams, bigrams, and trigrams) highlighted additional distinctions: fake news frequently contains emotionally charged or politically oriented wording, whereas real news relies on neutral, factual, and journalistic expressions.

These findings indicate that stylistic and structural cues play an important role in distinguishing real and fake content, which supports the use of a character-level CNN model capable of capturing such fine-grained patterns directly from raw text.

3.3.1 Text-Level Frequency Analysis

To further examine lexical differences between Fake and Real news, the most frequent unigrams and bigrams were analyzed after removing stopwords. These frequency distributions highlight the dominant topics and stylistic cues present in the datasets. As seen in the charts below, political entities and event-related expressions appear prominently, especially in Fake news samples.

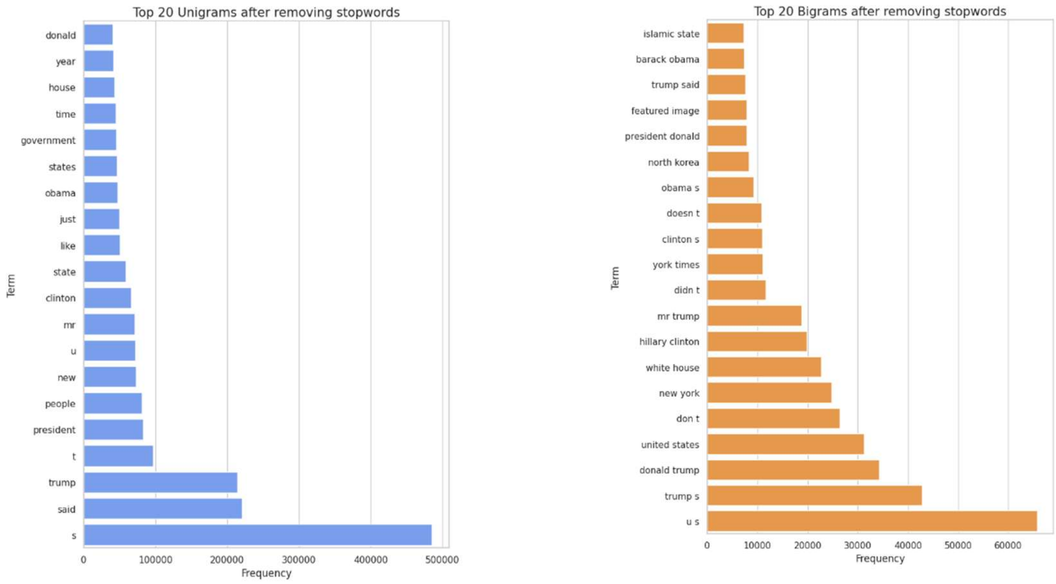

Figure 2-3. Top 20 Unigrams and Biagrams After Removing Stopwords

3.3.2 Sentiment and Length Characteristics

Real news articles are generally longer and exhibit more neutral sentiment, whereas Fake news shows slightly stronger negative polarity and emotional wording. These structural differences support the hypothesis that stylistic cues contribute to the effectiveness of character-level models.

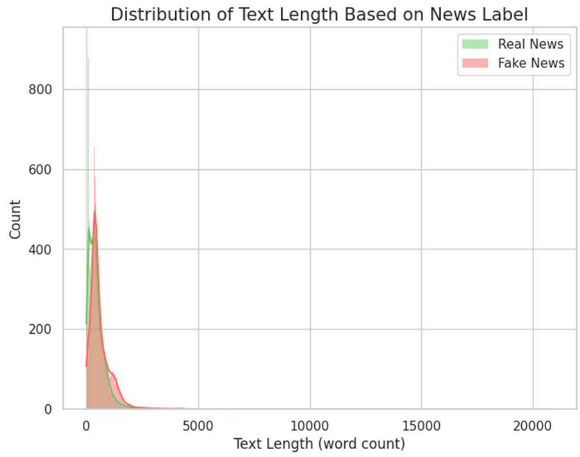

Figure 4. Distribution of Text Length

3.4 Models

3.4.1 Baseline Model

The baseline TF-IDF + Logistic Regression model achieved **92.8% accuracy** and **0.93 macro-F1**, demonstrating strong performance and balanced classification between Fake and Real news.

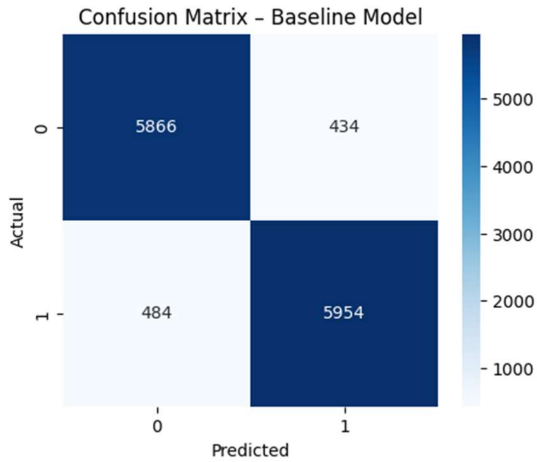

Figure 5. Baseline Confusion Matrix

3.4.2 Hyperparameter Optimization

Hyperparameter Optimization was applied to improve the baseline TF-IDF and Logistic Regression model. A RandomizedSearchCV procedure with 3-fold cross-validation was used, and macro-F1 was selected as the evaluation metric. Due to the size of the dataset, the search was performed on a stratified subset of 20,000 training samples, and the best parameter configuration was later retrained on the full training set. The optimal model used TF-IDF with unigrams and bigrams, a maximum document frequency of 0.95, a minimum document frequency of 2, and a vocabulary size of 80,000 features. The Logistic Regression component performed best with C = 2.0 and balanced class weights. After optimization, the model reached **94.2% accuracy** and a **0.94 macro-F1 score** on the test set, improving over the baseline performance and reducing misclassification errors for both fake and real news classes.

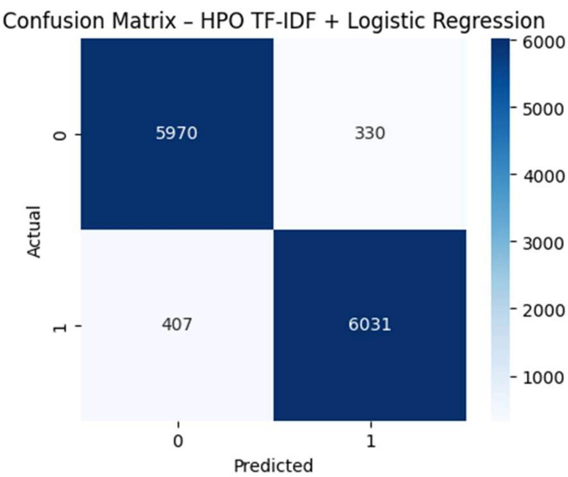

Figure 6. HPO and Logistic Regression Confusion Matrix

3.4.3 Character-level CNN

A character-level CNN was trained using 1,000-character sequences and a 95-character vocabulary. The model consists of an embedding layer, two convolution–pooling blocks, and a dense classifier. It achieved **97.9% accuracy** and a **macro-F1 of 0.98**, showing a clear improvement over both the baseline and the HPO-optimized TF-IDF model. The advanced model significantly reduced misclassification errors and captures fine-grained stylistic patterns useful for fake news detection.

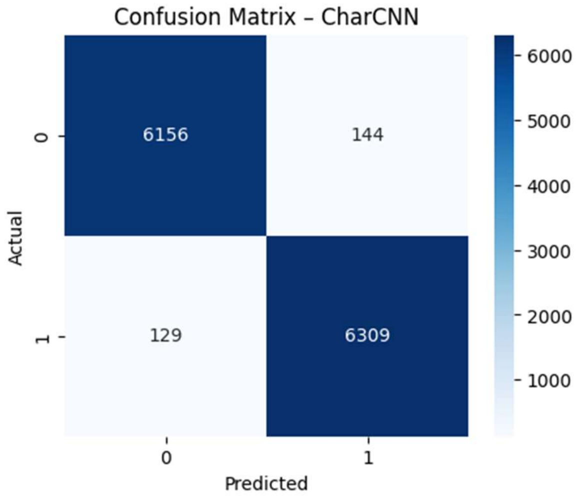

Figure 7. CharCNN Confusion Matrix

4. Results and Analysis

4.1 Overall Performance

Hyperparameter optimization further improved the TF-IDF + Logistic Regression model, increasing both accuracy and macro-F1 by selecting more effective feature and regularization settings. This demonstrates that even simple linear models benefit significantly from targeted optimization.

Table 3. Classification performance comparison of baseline, HPO and CharCNN models

Model	Accuracy	Macro-F1
Baseline (TF-IDF + LR)	0.9279	0.9279
HPO TF-IDF + LR	0.9421	0.9421
CharCNN (Character-level)	**0.9786**	**0.9785**

The results clearly indicate that the Character-level CNN substantially outperforms both the baseline and HPO-tuned TF-IDF models. While logistic regression provides strong lexical-level performance, the CharCNN captures deeper stylistic and structural patterns that are characteristic of fake news writing. This demonstrates that character-level modeling is highly effective for detecting subtle linguistic cues in misinformation.

4.2 Error Analysis

To better understand model behavior beyond aggregate performance metrics, an error analysis was conducted by examining error distributions, slice-based error rates, and representative misclassified examples.

Table 4. Distribution of error types

Error Type	Baseline	HPO
Correct	11,820	12,001
Real → Fake	484	407
Fake → Real	434	330

As shown in Table 4, both models exhibit a higher tendency to misclassify real news as fake rather than the reverse. This asymmetry suggests that stylistic cues commonly associated with sensational or politically framed writing may override factual correctness, particularly for word-based representations.

Table 5. Error rates by text length

Text Length	Baseline Error Rate	HPO Error Rate
Short	0.064	0.054
Medium	0.069	0.053
Long	0.083	0.066

Table 5 indicate that error rates increase with article length, suggesting that longer texts introduce additional structural and topical complexity. While HPO consistently improves performance across all slices, long articles remain the most challenging.

Table 6. Error rates by dataset

Dataset	Baseline Error Rate	HPO Error Rate
ISOT	0.031	0.025
WELFake	0.136	0.109

As shown Table 6 presents slice-based error rates by dataset source. Substantially higher error rates are observed on the WELFake dataset compared to ISOT, even after optimization. This performance gap highlights a domain shift effect and reduced generalization across heterogeneous news sources, emphasizing the importance of cross-dataset evaluation in fake news detection tasks.

Table 7. Representative misclassified news articles

True Label	Predicted Label	News Article (Excerpt)
Fake (0)	Real (1)	*Exodus from Puerto Rico could upend Florida vote in 2016 presidential race… It's important to vote and be heard — it's a privilege…*
Real (1)	Fake (0)	*How Donald Trump is already indirectly helping Russia and Syria… Fort Russ – PolitRussia – translated by J. Arnoldski…*
Real (1)	Fake (0)	*The Most Unhealthy Jobs in America… There are many jobs that are commonly understood to be dangerous to your health…*

To complement quantitative error analysis, representative misclassified samples were manually inspected in Table 7. These cases suggest that stylistic ambiguity and politically charged language can confuse lexical and character-level cues, leading to misclassification.

5. Discussion and Future Work

This study demonstrates that character-level modeling provides significant advantages for fake news detection by capturing fine-grained stylistic patterns that are difficult to represent using traditional word-based features. The strong performance of the Character-level CNN across both datasets confirms that stylistic cues such as capitalization, punctuation usage, and subword patterns play a crucial role in distinguishing fake and real news.

However, the error analysis reveals that domain shift remains an important challenge. Higher error rates on the WELFake dataset indicate that differences in source distribution and writing style can negatively affect generalization, even for deep learning models. Additionally, misclassifications observed in politically framed and stylistically ambiguous articles suggest that contextual understanding beyond surface-level text features is still limited.

Future work may explore hybrid architectures that combine character-level representations with contextual embeddings from transformer-based models to better capture semantic meaning. Expanding the study to multilingual datasets and incorporating temporal or source credibility features could further improve robustness. These directions may contribute to more reliable and scalable fake news detection systems.

6. Conclusion

In this study, the effectiveness of a Character-level Convolutional Neural Network for fake news detection was investigated and compared against a TF-IDF-based baseline and an optimized variant. Experimental results on the ISOT and WELFake datasets show that while hyperparameter optimization improves traditional models, the Character-level CNN achieves the highest accuracy and macro-F1 score.

The findings highlight that character-level representations are particularly effective in capturing stylistic and structural cues associated with deceptive content. Error analysis further confirms that such models reduce misclassification in challenging cases, although domain shift and stylistic ambiguity remain open challenges. Overall, this work demonstrates that character-level CNNs offer a robust and promising approach for fake news detection, providing a strong foundation for future research in misinformation analysis.

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